Mukul Kumar

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Education

Indian Institute of Technology Hyderabad — Aug 2022 – May 2026
Bachelor of Technology in Chemical Engineering — *GPA: 7.14 / 10*

R.K.C Vidya Vihar, Bohatwas Ahir (Rewari) — 2022
XII (CBSE) — *Percentage: 91.20%*

R.K.C Vidya Vihar, Bohatwas Ahir (Rewari) — 2020
X (CBSE) — *Percentage: 94.60%*

Work Experience

People Tech Group Inc — Jun – Aug 2025
Software Developer Intern – SDV (C++, gRPC, JSON, VSS, CMake) — *Hyderabad, India*

- Built and optimized C++ wrapper classes for Kuksa.val Databroker, improving Set/Get/Subscribe/Publish methods efficiency and reducing average latency by 15 ms across HVAC, Odometer & Tyre pressure signals.
- Integrated **gRPC & JSON APIs** enabling high-performance, real-time vehicle signal exchange, boosting inter-module throughput.
- Abstracted low-level APIs into reusable high-level libraries, cutting developer onboarding and integration time.
- Integration tested across multiple SDV modules, ensuring 99.9% system reliability on **Software Defined Vehicles**.

Projects

Splitwise Backend — **April – May 2026**
Java, Spring Boot, MySQL, Spring Data JPA, REST APIs — *GitHub*

- Built a backend expense-sharing platform using Spring Boot and MySQL, exposing REST APIs for expense tracking, balance computation, and debt settlement.
- Implemented a **Greedy Settlement Algorithm** reducing transaction count by nearly **40%**.
- Modeled many-to-many user-expense relationships using Spring Data JPA and normalized relational schemas.
- Structured application layers following service-repository architecture principles.

DocuMind AI — **July – Aug 2025**
Python, OpenAI Embeddings, Gemini, Docker — *GitHub*

- Built an AI-powered document intelligence platform supporting semantic search and retrieval.
- Designed a semantic retrieval pipeline using OpenAI embeddings and Google Gemini.
- Processed **10K+** document chunks while maintaining sub-second retrieval latency.
- Ranked among the **Top 100 teams** out of **60,000+ participants** in Adobe India Hackathon 2025.

Technical Skills

- **Languages:** C++, Java, Python, SQL
- **Backend:** Spring Boot, FastAPI, Node.js, Express.js, REST APIs
- **Databases:** MySQL, PostgreSQL
- **Distributed Systems:** gRPC, Protocol Buffers, WebSockets, SOME/IP
- **Developer Tools:** Docker, Linux, Git, GitHub, Postman
- **Core Concepts:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Achievements & Certifications

- **Competitive Programming:** Solved 450+ Data Structures & Algorithm problems across LeetCode and Codeforces.
- **LeetCode:** Achieved a peak rating of **1671**.
- **National Talent Search Examination (NTSE):** Qualified both Stage-I and Stage-II.
- **Sports:** Represented IIT Hyderabad in Volleyball at INTER IIT Sports Meet.